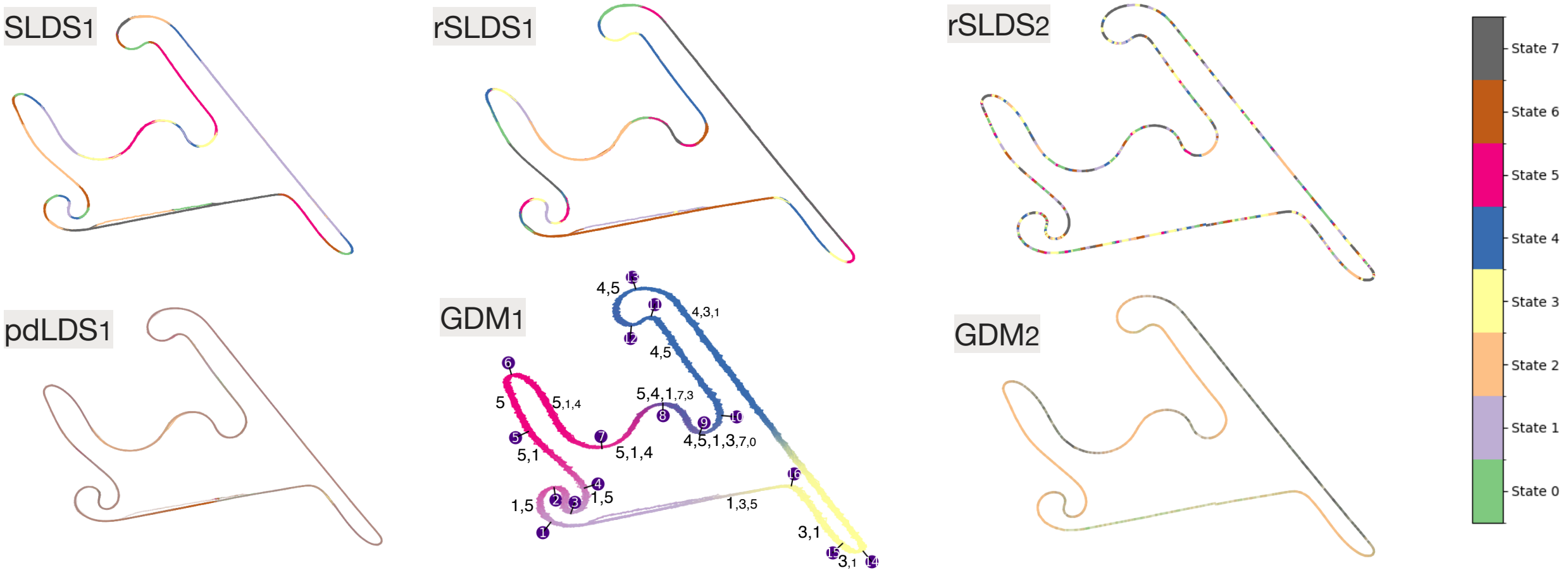

Supplementary Figure. Two versions of the F1 dataset with different input observations.
(1) **Position-based observations:** experiments on two tracks (Suzuka and Shanghai), where training and testing are performed on the same track using 1st- and 5th-place trajectories.
(2) **Velocity-based observations:** experiments on all 24 tracks, where the model is trained on 23 tracks and tested on a held-out track (Shanghai).
Inferred states are driven primarily by location in (1) and by motion dynamics in (2). The velocity-based setting yields consistent latent “velocity states” shared across circuits.